Elton Chang

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EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA
University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

RESEARCH INTERESTS

Spatial tactile representations, contact dynamics, and action-conditioned world models for physics simulation and robot learning.

INDEPENDENT RESEARCH

Tactile Prediction and Contact Mechanics Oct 2026 - Present

- Built a Python/PyTorch pipeline to predict vibration spectra from brief contacts in the Cluster Haptic Texture Dataset; compared retrieval, regression, and encoders across speeds, directions, and loads.
- Found retrieval outperformed the encoder in development tests; verified support-only inference and exactly reproduced 1,878 sensor windows and 22 prediction arrays.
- Implemented eight force-prediction methods and controls with surface-level holdouts and paired bootstrap intervals; evaluated 450 synthetic trials across 10 groups.
- Built analytical stopping and bounded release-speed calculations plus data ingestion with explicit units and force signs; physical validation remains pending.

PROJECTS

AMOS | Co-Founder 2026 - Present

- Developed a Rust control layer for an AI analyst that enforces tool permissions, executes SQL/statistics outside the LLM, and links claims to evidence for review and replay.

TraceFrame

- Developed a Python evidence tracker for pandas, Polars, and DuckDB workflows, adding dataset checks, lineage, and HTML audit reports to make analytical results inspectable.

Next Best Action (Collaboration with PNC)

- Compared HistGradientBoosting, LightGBM, XGBoost, and ensembles for next-month product adoption; achieved 0.811 ROC AUC and a 95% top-three hit rate on supported recommendations.

Quantifying Meritocracy (Research)

- Extended Talent vs. Luck with a solidarity parameter to compare redistribution strategies using inequality, mobility, and wealth concentration metrics in reproducible simulations.

EXPERIENCE

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA

- Improved XGBoost aircraft-on-ground risk ranking (Precision@20: 35% to 62%); tuned model parameters and thresholds with Optuna, increasing median warning lead time from 6.5 to 10.2 days.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA

- Standardized testing environments across 15+ non-production hosts with Terraform; built backup/replication telemetry and alerting, reducing anomaly detection time by 40%.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA

- Automated file ingestion, metadata indexing, and lifecycle tracking in a cloud-hosted file management system using scheduled Python jobs.

TECHNICAL SKILLS

AI Engineering: Python, Rust, SQL, PyTorch, scikit-learn, LLM tool orchestration, model evaluation
Robotics Research: Tactile signal processing, friction estimation, contact dynamics, NumPy/SciPy